

FAULT CODE 272 - Engine Fuel Pump Pressurizing Assembly 1 Circuit - Voltage Above Normal or Shorted to High Source

Warnings and Cautions

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

CAUTION

To reduce the possibility of pin and harness damage, use the following test lead when taking a measurement: Part Number 3822758 - male Deutsch™/AMP™/Metri-Pack™ test lead.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the fault codes.	
	STEP 1A. Check for an active fault code.	Fault Code 272 active or inactive with more than one count logged in the last 25 engine hours?
STEP 2.	Check the fuel pump actuator and circuit.	
	STEP 2A. Inspect the fuel pump actuator and connector pins.	Dirty or damaged pins?
	STEP 2B. Check the circuit response.	Fault Code 271 active and Fault Code 272 inactive?
	STEP 2C. Check the fault codes and verify fuel pump actuator condition.	Fault Code 272 active?

STEPS		SPECIFICATIONS
STEP 3.	Check the engine control module (ECM) and engine harness.	
	STEP 3A. Inspect the ECM and engine harness connector pins.	Dirty or damaged pins?
	STEP 3B. Check the ECM response.	Fault Code 271 active and Fault Code 272 inactive?
	STEP 3C. Check for an open circuit in the engine harness.	Less than 10 ohms?
	STEP 3D. Check for a pin-to-pin short circuit in the engine harness.	Greater than 100k ohms?
	STEP 3E. Check for an inactive fault code.	Fault Code 272 inactive?
STEP 4.	Check ECM calibration and clear fault codes.	
	STEP 4A. Check if an ECM calibration update is available.	If a calibration update for this fault code is available, does the ECM contain that revision or higher?
	STEP 4B. Disable the fault code.	Fault code inactive?

STEP 1. Check the fault codes.

STEP 1A. Check for an active fault code.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Check for an active fault code. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 272 active or inactive with more than one count logged in the last 25 engine hours? YES	2A
	Fault Code 272 active or inactive with more than one count logged in the last 25 engine hours? NO	Use the following procedure for an inactive or intermittent fault code. Refer to Procedure 019-362 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html)

STEP 2. Check the fuel pump actuator and circuit.

STEP 2A. Inspect the engine harness and actuator connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the fuel pump actuator from the engine harness.

Action	Specification/Repair	Next Step
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Inspect the engine harness and actuator connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the actuator connector or engine harness connector. Clean the connector and pins. Repair the damaged harness, connector, or pins, if possible. Refer to Procedure 019-043 in the Associated Procedures Table.	4A
	Dirty or damaged pins? NO	2B

STEP 2B. Check the circuit response.

Conditions : • Turn keyswitch OFF. • Wait 30 seconds. • Disconnect the fuel pump actuator from the engine harness.		
Action	Specification/Repair	Next Step

Check the circuit response. - Place a jumper wire between the fuel pump actuator SIGNAL pin and the fuel pump actuator RETURN pin at the fuel pump actuator connector of the engine harness. - Turn keyswitch ON. - Connect INSITE™ electronic service tool. Use the circuit diagram or the wiring diagram for connector pin identification. Check for the appropriate circuit response after 30 seconds. - Use INSITE™ electronic service tool to read the fault codes.	Fault Code 271 active and Fault Code 272 inactive? YES	2C
	Fault Code 271 active and Fault Code 272 inactive? NO	3A

STEP 2C. Check the fault codes and verify fuel pump actuator condition.

Conditions :

- Turn keyswitch OFF.
- Connect the fuel pump actuator to the engine harness.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Check for the appropriate circuit response after 30 seconds. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 272 active? YES Repair : A damaged fuel pump actuator has been detected. Replace the fuel pump actuator. Refer to Procedure 019-117 in the Associated Procedures Table.	4A
	Fault Code 272 active? NO Repair : None. The removal and installation of the connector corrected the fault.	4A

STEP 3. Check the ECM and engine harness.

STEP 3A. Inspect the ECM and engine harness connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness from the ECM connector.

Action	Specification/Repair	Next Step
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Inspect the engine harness and ECM connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the ECM connector or original equipment manufacturer (OEM) harness connector. Clean the connector and pins. Repair the damaged harness, connectors, or pins, if possible. Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)	4A
	Dirty or damaged pins? NO	3B

STEP 3B. Check the ECM response.

Conditions : • Turn keyswitch OFF. • Disconnect the engine harness from the ECM connector.		
Action	Specification/Repair	Next Step

Check the circuit response. - Place a jumper wire between the fuel pump actuator SIGNAL pin and the fuel pump actuator RETURN at the ECM engine connector. - Turn keyswitch ON. - Connect INSITE™ electronic service tool. Use the circuit diagram or the wiring diagram for connector pin identification. Check for the appropriate circuit response after 30 seconds. - Use INSITE™ electronic service tool to read the fault codes.	Fault Code 271 active and Fault Code 272 inactive? YES	3C
	Fault Code 271 active and Fault Code 272 inactive? NO Repair : Replace the ECM. Refer to Procedure 019-031 in the Associated Procedures Table.	4A

STEP 3C. Check for an open circuit in the engine harness.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness from the ECM connector.
- Disconnect the fuel pump actuator from the engine harness.

Action	Specification/Repair	Next Step

Check for an open circuit. • Measure the resistance between the fuel pump actuator SIGNAL pin at the ECM connector of the engine harness and the fuel pump actuator SIGNAL pin at the fuel pump actuator connector of the engine harness. • Measure the resistance between the fuel pump actuator RETURN pin at the ECM connector of the engine harness and the fuel pump actuator RETURN pin at the fuel pump actuator connector of the engine harness. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Less than 10 ohms? YES	3D
	Less than 10 ohms? NO Repair : Repair or replace the engine harness. Refer to Procedure 019-043 in the Associated Procedures Table.	4A

STEP 3D. Check for a pin-to-pin short circuit in the engine harness.

Conditions : • Turn keyswitch OFF. • Disconnect the engine harness from the ECM connector. • Disconnect the fuel pump actuator from the engine harness.		
Action	Specification/Repair	Next Step

Check for a pin-to-pin short circuit on one connector. Measure the resistance between the fuel pump actuator SIGNAL pin in the engine harness ECM connector and all other pins in the connector. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? YES	3E
	Greater than 100k ohms? NO Repair : Repair or replace the engine harness. Refer to Procedure 019-043 in the Associated Procedures Table.	4A

STEP 3E. Check for an inactive fault code.

Conditions : - Connect all components. - Turn keyswitch ON. - Connect INSITE™ electronic service tool.		
Action	Specification/Repair	Next Step

Check for the appropriate circuit response after 30 seconds. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 272 inactive? YES Repair : None. The removal and installation of the connector corrected the fault.	4A
	Fault Code 272 inactive? NO Repair : Troubleshooting procedures need to be repeated from the beginning. A failure mode should have been detected.	1A

STEP 4. Check ECM calibration and clear fault codes.

STEP 4A. Check if an ECM calibration update is available.

Conditions :

- Connect all components.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step

Compare the ECM code and revision number in the ECM to the calibration revisions listed in the ECM Calibration Revision History for applicable changes related to this fault code. Use INSITE™ electronic service tool to find the present ECM code and revision number in the ECM. The ECM code and revision number are found in the Calibration Information section of System ID and Dataplate in Features and Parameters.	If a calibration update for this fault code is available, does the ECM contain that revision or higher? YES	4B
	If a calibration update for this fault code is available, does the ECM contain that revision or higher? NO Repair : If necessary, calibrate the ECM. Refer to Procedure 019-032 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-032.html)	4B

STEP 4B. Disable the fault code.

Conditions : - Connect all components. - Connect INSITE™ electronic service tool.		
Action	Specification/Repair	Next Step
Disable and clear the fault code. Operate the engine within the "Conditions for Clearing the Fault Code" found in the Overview section of this troubleshooting procedure.	Fault code inactive? YES	Repair complete.
	Fault code inactive? NO Repair : Return to the troubleshooting steps or contact a Cummins® Authorized Repair Location if all steps have been completed and checked again.	1A

Associated Procedures

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/164/164-019-031.html)	ISL9.5 CM2150 SN	4310608
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/218/218-019-031.html)	QSB4.5 CM2150 B108	4326163
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/216/216-019-031.html)	QSB6.7 CM2150 B109	4326168
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/331/331-019-031.html)	QSB7 CM2880 B117	4358390
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/177/177-019-031-tr.html)	ISG11 CM2880 G106	4332695
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/177/177-019-031-tr.html)	ISG12 CM2880 G107	4332690
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/325/325-019-031-tr.html)	ISG11 CM2880 G108	4332901

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/325/325-019-031-tr.html)	ISG12 CM2880 G109	4332906
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/284/284-019-031-tr.html)	QSF2.8 CM2880 F104	4332741
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/284/284-019-031-tr.html)	QSF2.8 CM2880 F108	4332746
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/336/336-019-031.html)	ISL8.9 CM2880 L112	4358493
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/389/389-019-031-tr.html)	QSF3.8 CM2880 F112	4383825
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/403/403-019-031.html)	QSL9.3 CM2880 L113	4383811
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/379/379-019-031.html)	ISB/ISD6.7 CM2880 B126	4383693
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/506/506-019-031.html)	ISD6.7 CM2880 D101	5411372

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/497/497-019-031.html)	ISF2.8 CM2220 F129	5411325
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/501/501-019-031.html)	ISF4.5 CM2220 F123	5411320
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/417/417-019-031.html)	QSG12 CM2880 G112	4388731
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/556/556-019-031.html)	ISF3.8 CM2220 F134B	5504165
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/577/577-019-031.html)	Z14 CM2670 Z103B	5504577
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/546/546-019-031.html)	D6.7 CM2670 D102B	5504515
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/629/629-019-031.html)	B6.2 CM2670 B156B	5579510
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/624/624-019-031.html)	X12 CM2670 X121B	5504455

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/622/622-019-031.html)	L9 CM2670 L128B	5504589
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/164/164-019-043.html)	ISL9.5 CM2150 SN	4310608
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/218/218-019-043.html)	QSB4.5 CM2150 B108	4326163
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/216/216-019-043.html)	QSB6.7 CM2150 B109	4326168
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/331/331-019-043.html)	QSB7 CM2880 B117	4358390
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/177/177-019-043-tr.html)	ISG11 CM2880 G106	4332695
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/177/177-019-043-tr.html)	ISG12 CM2880 G107	4332690
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/325/325-019-043-tr.html)	ISG11 CM2880 G108	4332901

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/325/325-019-043-tr.html)	ISG12 CM2880 G109	4332906
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/284/284-019-043.html)	QSF2.8 CM2880 F104	4332741
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/284/284-019-043.html)	QSF2.8 CM2880 F108	4332746
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/336/336-019-043.html)	ISL8.9 CM2880 L112	4358493
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/389/389-019-043.html)	QSF3.8 CM2880 F112	4383825
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/403/403-019-043.html)	QSL9.3 CM2880 L113	4383811
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/379/379-019-043.html)	ISB/ISD6.7 CM2880 B126	4383693
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/506/506-019-043.html)	ISD6.7 CM2880 D101	5411372

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/497/497-019-043.html)	ISF2.8 CM2220 F129	5411325
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/501/501-019-043.html)	ISF4.5 CM2220 F123	5411320
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/417/417-019-043.html)	QSG12 CM2880 G112	4388731
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/556/556-019-043.html)	ISF3.8 CM2220 F134B	5504165
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/577/577-019-043.html)	Z14 CM2670 Z103B	5504577
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/546/546-019-043.html)	D6.7 CM2670 D102B	5504515
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/629/629-019-043.html)	B6.2 CM2670 B156B	5579510
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/624/624-019-043.html)	X12 CM2670 X121B	5504455

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/622/622-019-043.html)	L9 CM2670 L128B	5504589
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/164/164-019-117.html)	ISL9.5 CM2150 SN	4310608
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/218/218-019-117.html)	QSB4.5 CM2150 B108	4326163
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/216/216-019-117.html)	QSB6.7 CM2150 B109	4326168
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/331/331-019-117.html)	QSB7 CM2880 B117	4358390
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/177/177-019-117.html)	ISG11 CM2880 G106	4332695
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/177/177-019-117.html)	ISG12 CM2880 G107	4332690
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/325/325-019-117.html)	ISG11 CM2880 G108	4332901

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/325/325-019-117.html)	ISG12 CM2880 G109	4332906
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/284/284-019-117.html)	QSF2.8 CM2880 F104	4332741
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/284/284-019-117.html)	QSF2.8 CM2880 F108	4332746
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/336/336-019-117.html)	ISL8.9 CM2880 L112	4358493
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/389/389-019-117.html)	QSF3.8 CM2880 F112	4383825
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/403/403-019-117.html)	QSL9.3 CM2880 L113	4383811
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/379/379-019-117.html)	ISB/ISD6.7 CM2880 B126	4383693
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/506/506-019-117.html)	ISD6.7 CM2880 D101	5411372

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/497/497-019-117.html)	ISF2.8 CM2220 F129	5411325
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/501/501-019-117.html)	ISF4.5 CM2220 F123	5411320
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/417/417-019-117.html)	QSG12 CM2880 G112	4388731
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/556/556-019-117.html)	ISF3.8 CM2220 F134B	5504165
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/577/577-019-117.html)	Z14 CM2670 Z103B	5504577
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/546/546-019-117.html)	D6.7 CM2670 D102B	5504515
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/629/629-019-117.html)	B6.2 CM2670 B156B	5579510
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/624/624-019-117.html)	X12 CM2670 X121B	5504455

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Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Fuel Pump Actuator	Refer to Procedure 019-117 (/qs3/pubsys2/xml/en/procedures/622/622-019-117.html)	L9 CM2670 L128B	5504589

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